



Computer Science and Engineering

Final Thesis Presentation

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Quantum Optimization for Node Placement in Device-Free Localization

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The American University in Cairo

Previous Committee questions

- **Q1: What is the key difference between Paper 1 and Paper 2? → [Slide 3](#)**
- **Q2: What is the coverage metric? → [Slide 7 & 8](#)**
- **Q3: How does the grid resolution affect the system? → [Slide 7](#)**
- **Q4: What is the transmission range assumption, and does the dataset satisfy it? → [Slide 9](#)**

Device Free Localization (DFL) VS Device Based Localization(DBL)

Offline Phase

- A human walks through the area carrying a device
- RSS fingerprints are collected at known reference locations
- A fingerprint database is built: location → RSS vector
- This phase is done once and stored

Online Phase

	DBL	DFL
Who has a device?	The tracked user	Nobody
What generates the signal?	User's own phone/tag	APs and MPs
What changes in RSS?	User's device readings	user's body
Who is trackable?	Only device carriers	Anyone

Table 1: Difference between DBL and DFL in the Online Phase

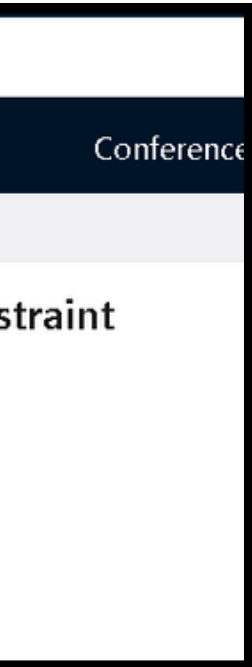
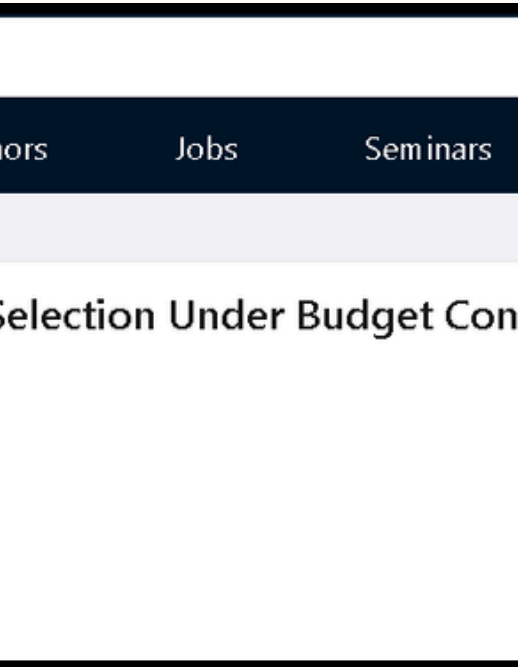
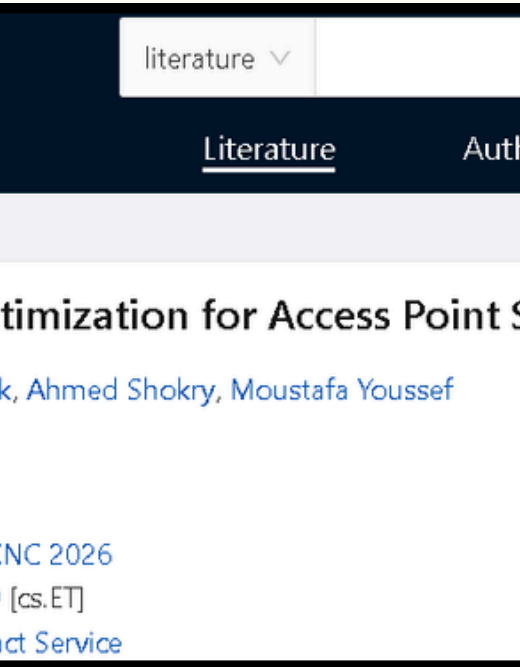
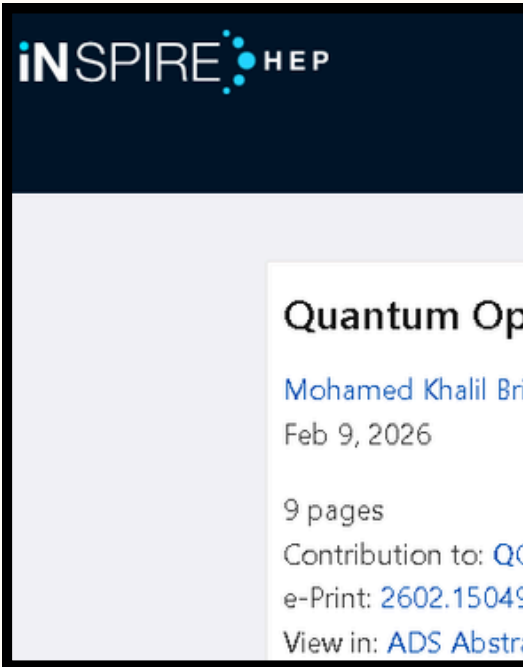
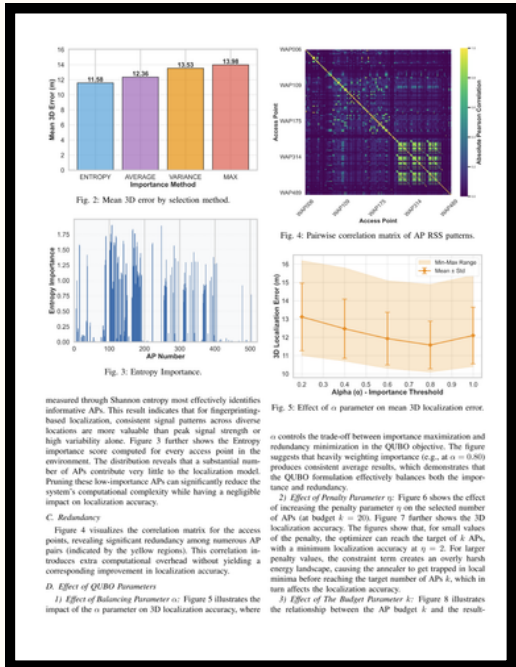
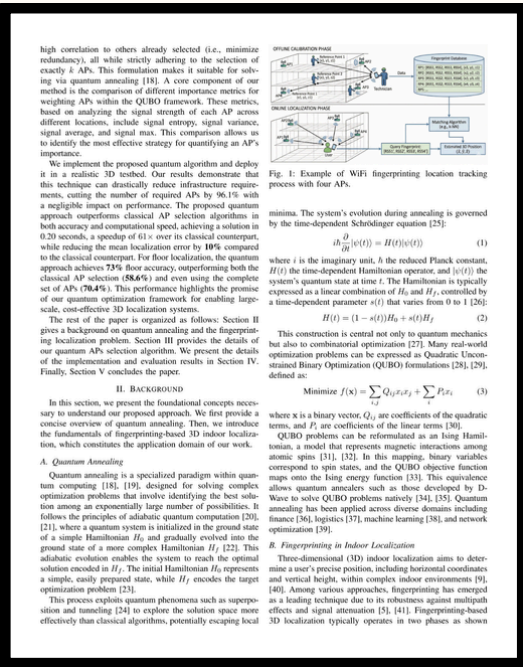
Same Topic But Different Problem (Q1)

	Quantum Optimization for Access Point Selection [Thesis 1]	Quantum Optimization for Node Placement in Device-Free Localization [Thesis 2]
Problem	Which WiFi access points should a device-based indoor localization system use?	Which AP-MP node pairs should a device-free localization system activate?
Setting	User carries a phone. Select k APs from a pool of 520 candidates.	No device carried. Select directed AP-MP links under two separate budgets (n APs and m MPs)
Constraint	Single budget on total AP count	Two separate budgets: one on AP count and one on MP count.
Key result	96.1% AP reduction, $61\times$ speedup over SA, best 3D localization accuracy	Lower mean error than SA (2.66 vs 2.69 m), $129\times$ speedup

Table 2: Comparison of the two thesis contributions — same quantum optimization framework applied to different localization paradigms

Our previous Work: 1 Paper Accepted & Presented

- Presented at the IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2026), Japan.
- **The Problem:** Dense deployments of WiFi Access Points (APs) in modern buildings create signal redundancy, increase computational complexity, and inflate tracking costs.
- **The Solution:** We formulated a budget-constrained quantum algorithm that selects the near-optimal subset of APs by maximizing signal importance and minimizing data overlap.
- **96.1% AP Reduction | 61x Speedup. | 73% Floor Accuracy**



Source: Authors

Source: Screenshot from [Link](#)

Why Device-Free Localization?

- Tracking people's positions traditionally requires to carry a device
- Many real scenarios make this impractical or impossible
 - Elderly residents who may forget or refuse to wear a tracker
 - Intruders who obviously will not cooperate
 - Building occupancy monitoring at scale
- Device-free localization (DFL) solves this by using RF signals
- In RTI, sensor nodes form a mesh of RF links
- A person's body attenuates any link whose Fresnel zone it crosses
→ those RSS drops reveal their position
- **Fresnel Zone**: The first Fresnel zone of link (A,B) is the set of all points P satisfying $d(P,A) + d(P,B) \leq d(A,B) + \lambda/2$, an ellipsoid with foci at A and B. Any object inside it causes measurable RF attenuation. (λ is the wavelength of the RF signal)

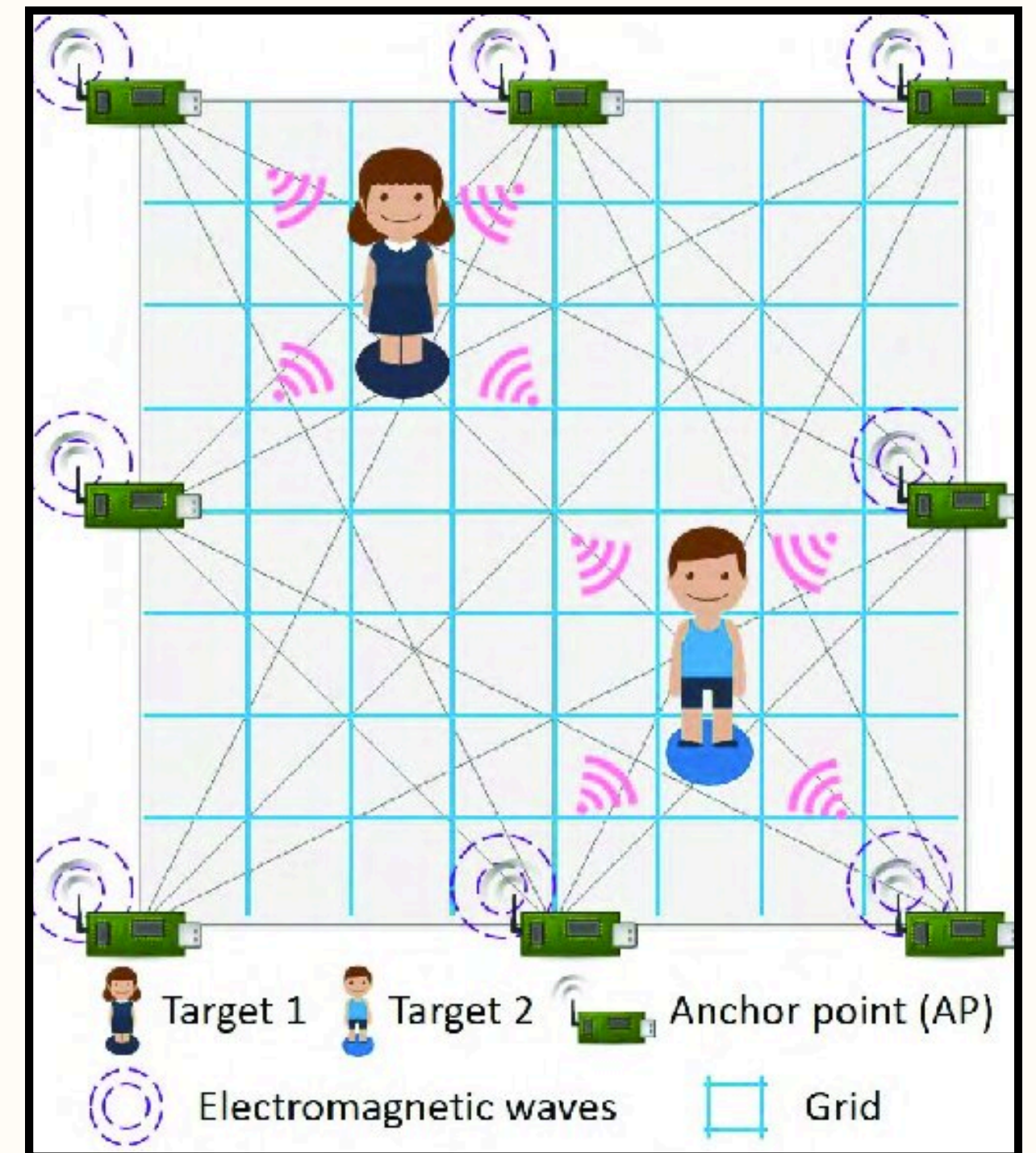


Figure 1: Multi-target device-free localization using electromagnetic waves and anchor points

Source: H. Huang et al., "An Accurate and Efficient Device-Free Localization Approach Based on Sparse Coding in Subspace," IEEE Access, vol. 6, pp. 61782–61799, 2018.

The Problem: Which Nodes Do You Activate?

- A DFL system has a pool of N candidate node positions
- Each node acts as either a transmitter (AP) or a receiver (MP)
- Budget limits how many APs (n) and MPs (m) can be active
- A link that crosses the monitored area at a useful angle is informative: its RSS changes a lot depending on where the person is
- Two links that run parallel and respond identically to movement are redundant: selecting both wastes budget

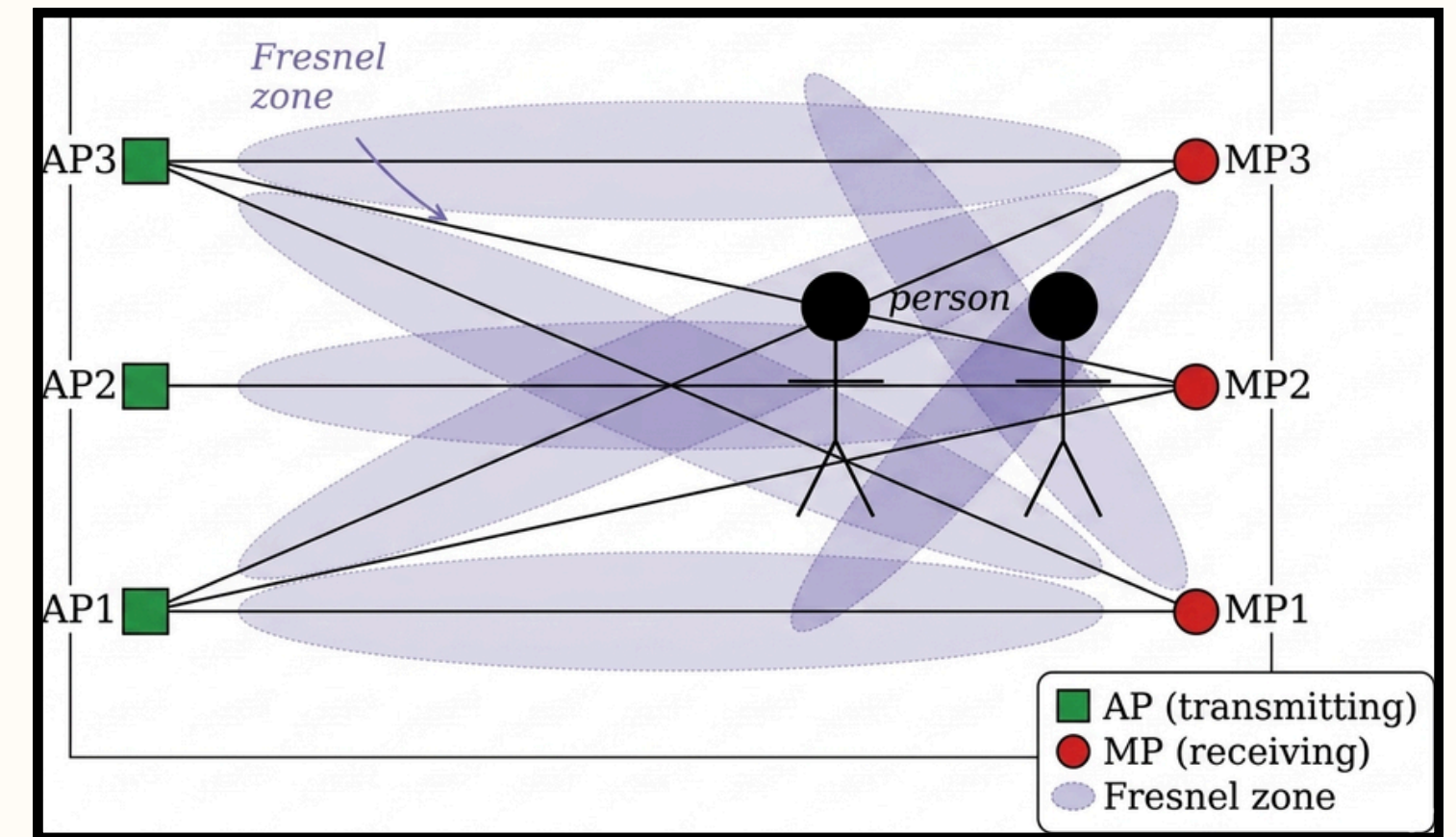


Figure 2: RTI deployment showing AP nodes, MP nodes, directed links, and Fresnel zone ellipses

Source: Diagram generated with Gemini (Google), 2026.

The Problem: Which Nodes Do You Activate?

- Good placement maximizes both individual link quality and collective spatial diversity
- Coverage is measured by the number of grid points in the monitored area that fall within at least one active link's Fresnel zone → A deployment with high coverage ensures the system can detect movement anywhere in the area, not just near the deployed nodes **(Q2)**.

- You choose ONE parameter (e.g. N points per axis); spacing follows automatically: $\text{spacing} = \text{room width} / (N-1)$. **(Q3)**.
- 1 grid cell gives no spatial discrimination
- A finer grid (more points, smaller spacing) increases runtime only, because more coverage checks are performed per candidate deployment

The Problem: Which Nodes Do You Activate?

Deployment Coverage: Same Budget (5 Active Links) — Three Configurations

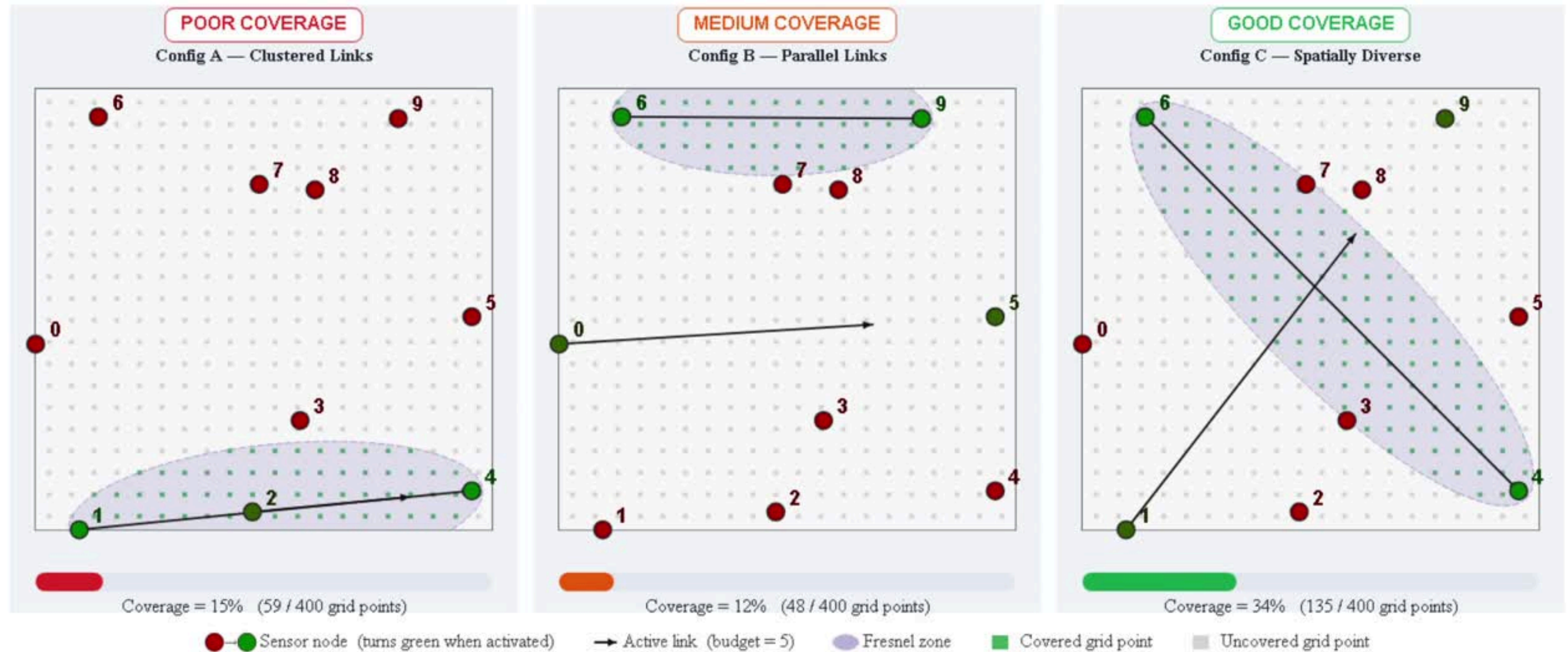


Figure 3: Different Activated Nodes Animation

Source: HTML code generated with Claude

Why Is This Hard?

- The AP-MP placement problem is a combinatorial subset selection problem with an exponential search space
- Search space = $C(N, n) \times C(N, m) \rightarrow$ choosing n APs and m MPs independently from N candidates
- For $n = m = N/2$: $C(N, N/2)^2 \approx O(4^N / N) \rightarrow$ exponential in N
- At $N=20$, $n=m=10$: $C(20,10)^2 = 184,756^2 \approx 34$ billion candidates
- The problem has two separate budget constraints: one on AP count, one on MP count
 - The standard fix in QUBO is to add a penalty term η .
 - Too small: the solver ignores the budget entirely
 - Too large: the penalty dominates, and the solver picks feasible but useless deployments
- η must be re-tuned for every budget and environment

Our Approach

- Cast AP-MP node selection as a QUBO
- The QUBO objective uses two link-level metrics computed directly from RSS fingerprint data:
 - **Importance:** how much links RSS varies across fingerprint locations: high variance means the link is sensitive to movement
 - **Redundancy:** absolute Pearson correlation between links p and $q \rightarrow$ high correlation means they carry the same information
- $\alpha \in [0,1]$ trades importance against redundancy
- Budget constraints are enforced through a provably correct binary search over α

We build our fingerprint DB in P time. (Use 1 AP and 1 MP) $\rightarrow O(K \cdot N^2)$

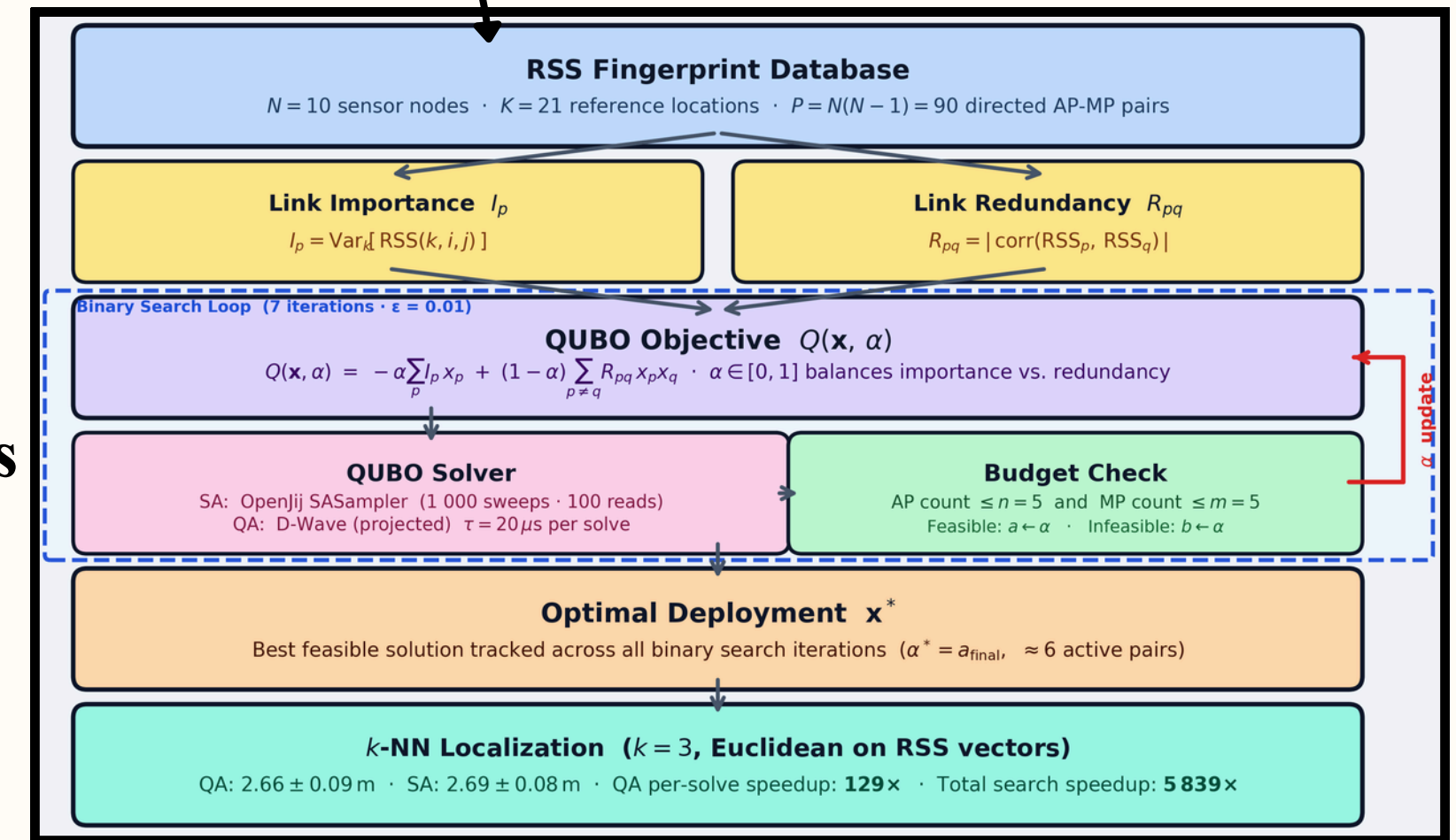


Figure 4: End-to-end pipeline from RSS fingerprint database to optimal AP-MP deployment

Source: Authors

The QUBO Objective

Define a single-parameter objective $Q(\mathbf{x}, \alpha)$ that trades off importance versus redundancy:

$$Q(\mathbf{x}, \alpha) = -\alpha \sum_p I_p x_p + (1 - \alpha) \sum_{p \neq q} R_{pq} x_p x_q$$

Figure 5: QUBO Hamiltonian formulation balancing link importance against inter-link redundancy *Source: Authors*

- First term is [negative](#): minimizing Q rewards $\rightarrow$ selecting high-importance links
- Second term is [positive](#): minimizing Q penalizes selecting two correlated links together
- When $\alpha=1$: pure [importance](#) maximization $\rightarrow$ selects all possible links, including redundant ones
- When $\alpha=0$: pure [redundancy](#) minimization $\rightarrow$ selects NO links, any selection increases redundancy
- The size of $\mathbf{x}$ is N^2
- Each binary variable $x_p \in \{0, 1\}$ represents whether directed AP-MP pair p is active in the deployment.
 - Selecting $x_p = 1$ means node i transmits and node j receives on that link.
 - Enabling link $(i \rightarrow j)$ does NOT automatically enable $(j \rightarrow i)$.
 - The budget checks how many unique nodes appear among selected links.

Budget Enforcement via Binary Search

- The QUBO objective has no budget term $\rightarrow$ so how do we enforce the AP and MP limits?
- Key insight: as α increases, the unconstrained optimum activates more nodes and we proved this is monotone (Theorem 1)
- Because node counts are non-decreasing in α , the set of feasible α values forms a contiguous interval $[0, \alpha_{\max}]$
- This means binary search is guaranteed to work $\rightarrow$ there are no gaps or reversals to trap it
- The algorithm maintains two bounds: a (always feasible) and b (always infeasible)
- At each step: solve the QUBO at the midpoint $\alpha = (a+b)/2$
 - If AP count $\leq n$ and MP count $\leq m \rightarrow$ raise a, record the localization error
 - If over budget $\rightarrow$ lower b

Budget Enforcement via Binary Search

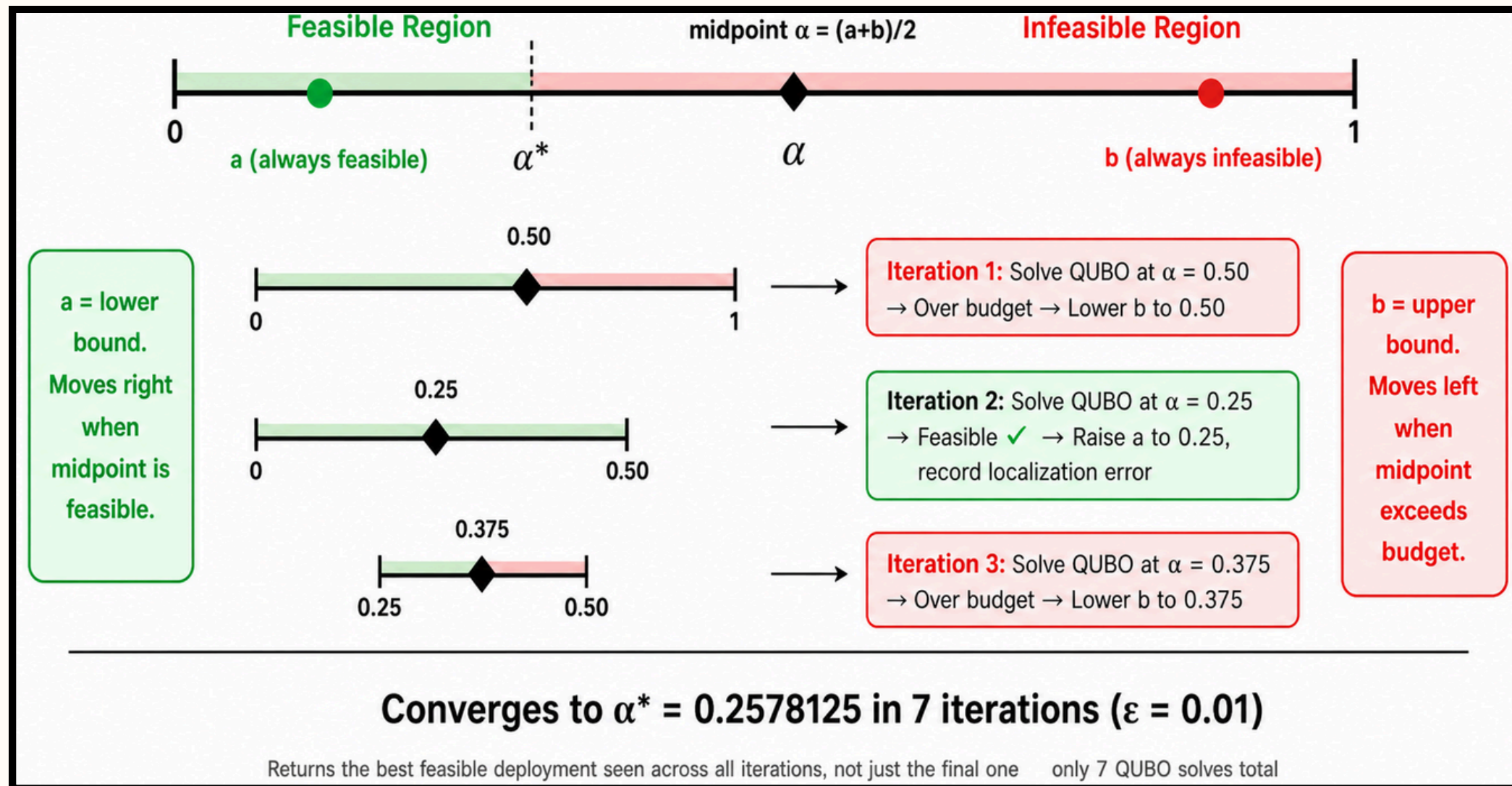


Figure 6: Binary search over α converging to the optimal feasible deployment in 7 iterations.

Source: Diagram generated with Gemini (Google), 2026.

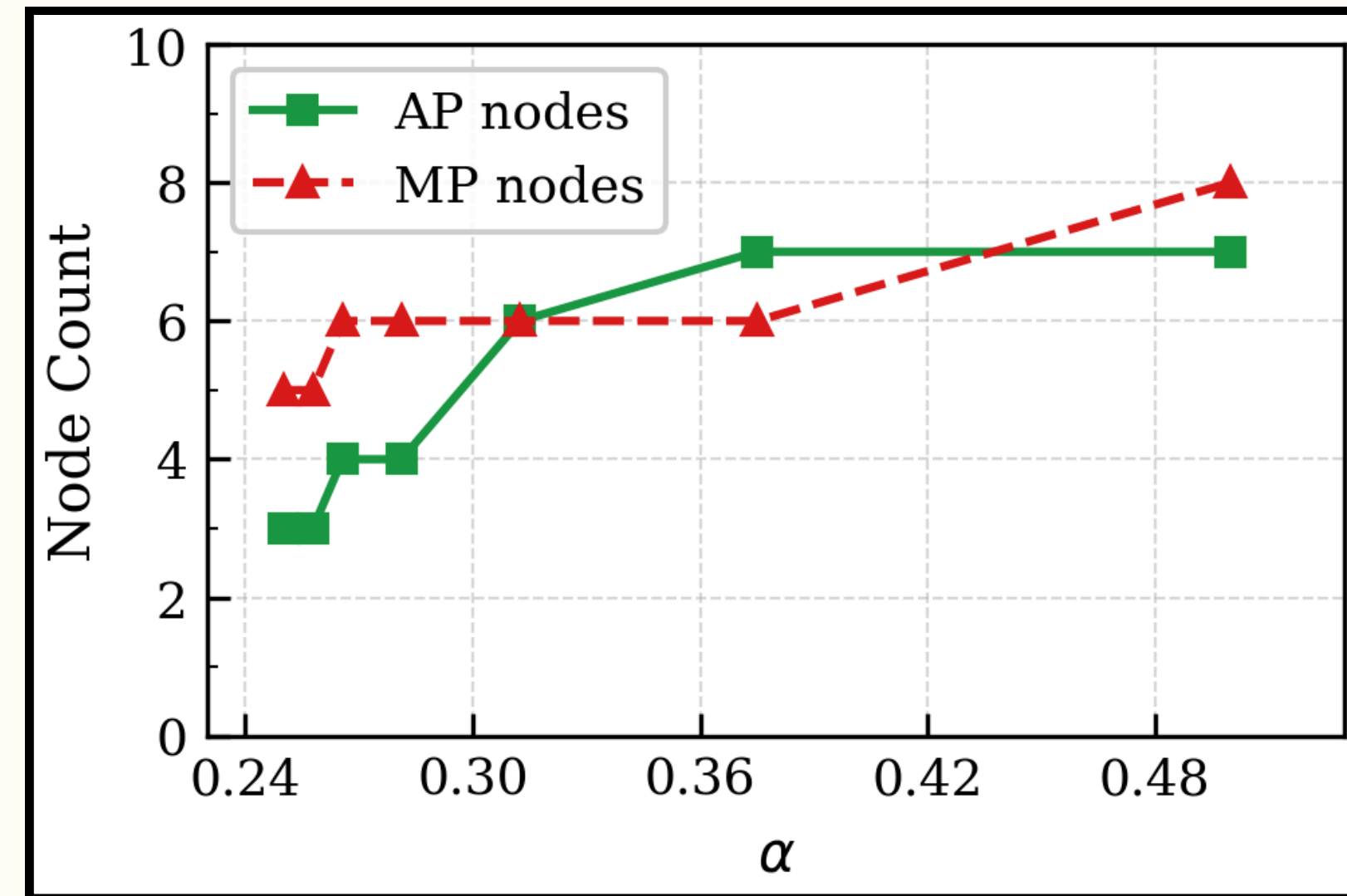


Figure 7: AP and MP node counts as a function of α

Source: Authors

Experimental Setup

Dataset

RTI Indoor Basement, N=10, K=21 fingerprint locations, P=90 pairs

Budget

n=m=5 APs and MPs

Two solvers

SA (measured, 1000 sweeps, 100 reads) vs QA (Simulated QA)

Evaluation

KNN classifier (k=3), 100 independent runs

Metric

localization error in meters + Time to Solution 99

Assumption: Full Network Connectivity (Q4)

We exclude putting an AP and an MP in the same location.

- All $N(N-1) = 90$ directed links produce a measurable RSS reading at every reference location

Distance between $P(0,0)$ and $P(8.5, 8.2) = 11.8$ m

- Room is 8.5×8.2 m $\rightarrow$ maximum node separation ≈ 11.8 m

- 2.4 GHz WiFi indoor range: 20–40 m $\rightarrow$ all 90 links are within range

- No link is filtered or discarded before the QUBO $\rightarrow$ operates on the complete candidate graph

- The budget constraint (≤ 5 APs, ≤ 5 MPs) is the only filter

Results: Localization Accuracy & Budget Constraint

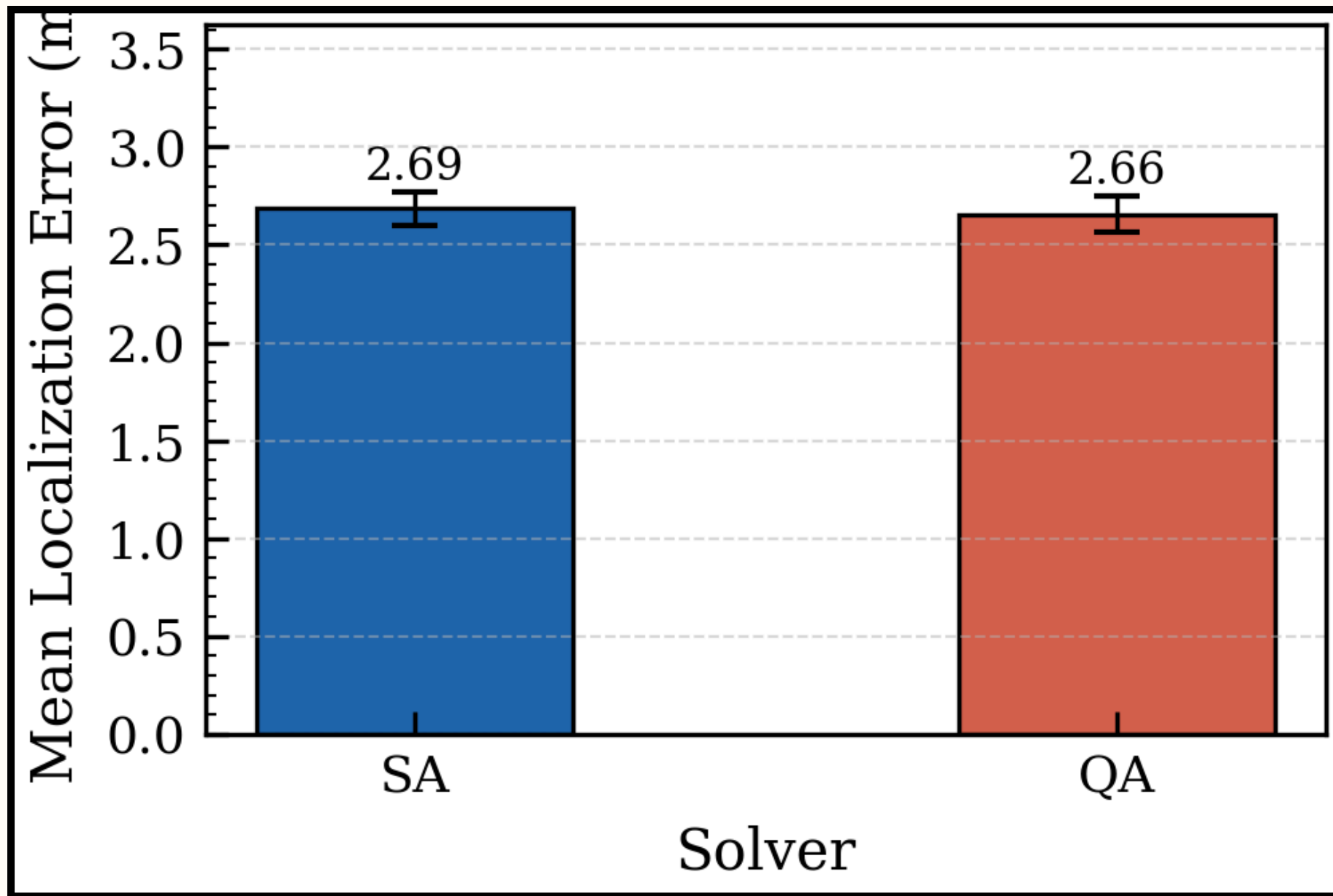


Figure 8: Mean localization error ($\pm$ one standard deviation) over 100 independent runs for SA and QA

Source: Authors

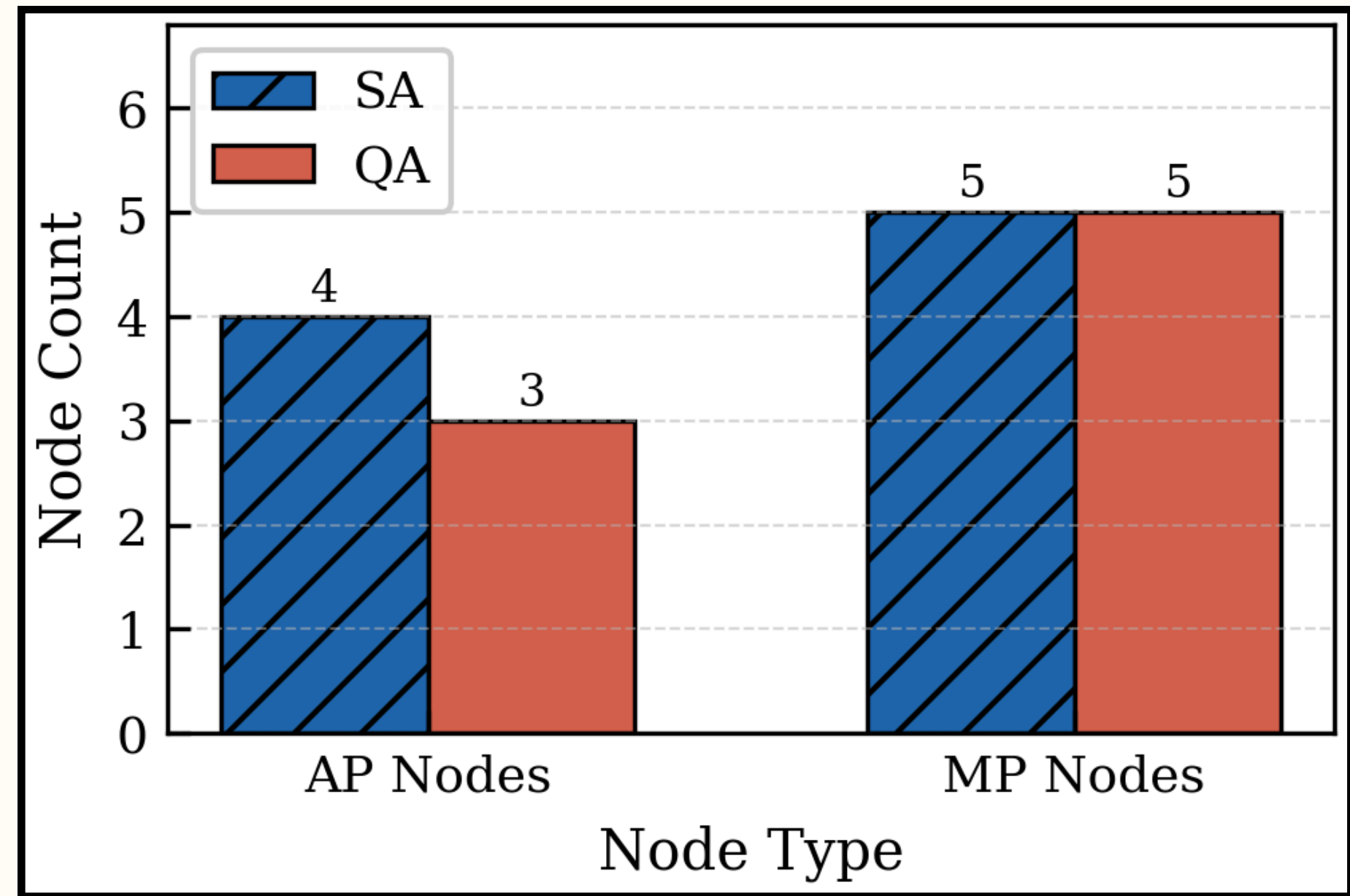


Figure 9: Active AP and MP node counts at α for SA and QA — neither solver saturates the budget*

Source: Authors

Results: Quantum TTS

1. **Run solver once (time = τ)**
2. **Record best solution found**
3. **Is this the best seen across all runs so far?**
4. **After all r runs: return the overall best**

ps = fraction of runs that matched the overall best solution found.

- **TTS99 measures the time needed find the optimal solution with 99% probability across all runs**
- **Both solvers achieved ps=1.0 across all runs**
- **129× in favor of QA**

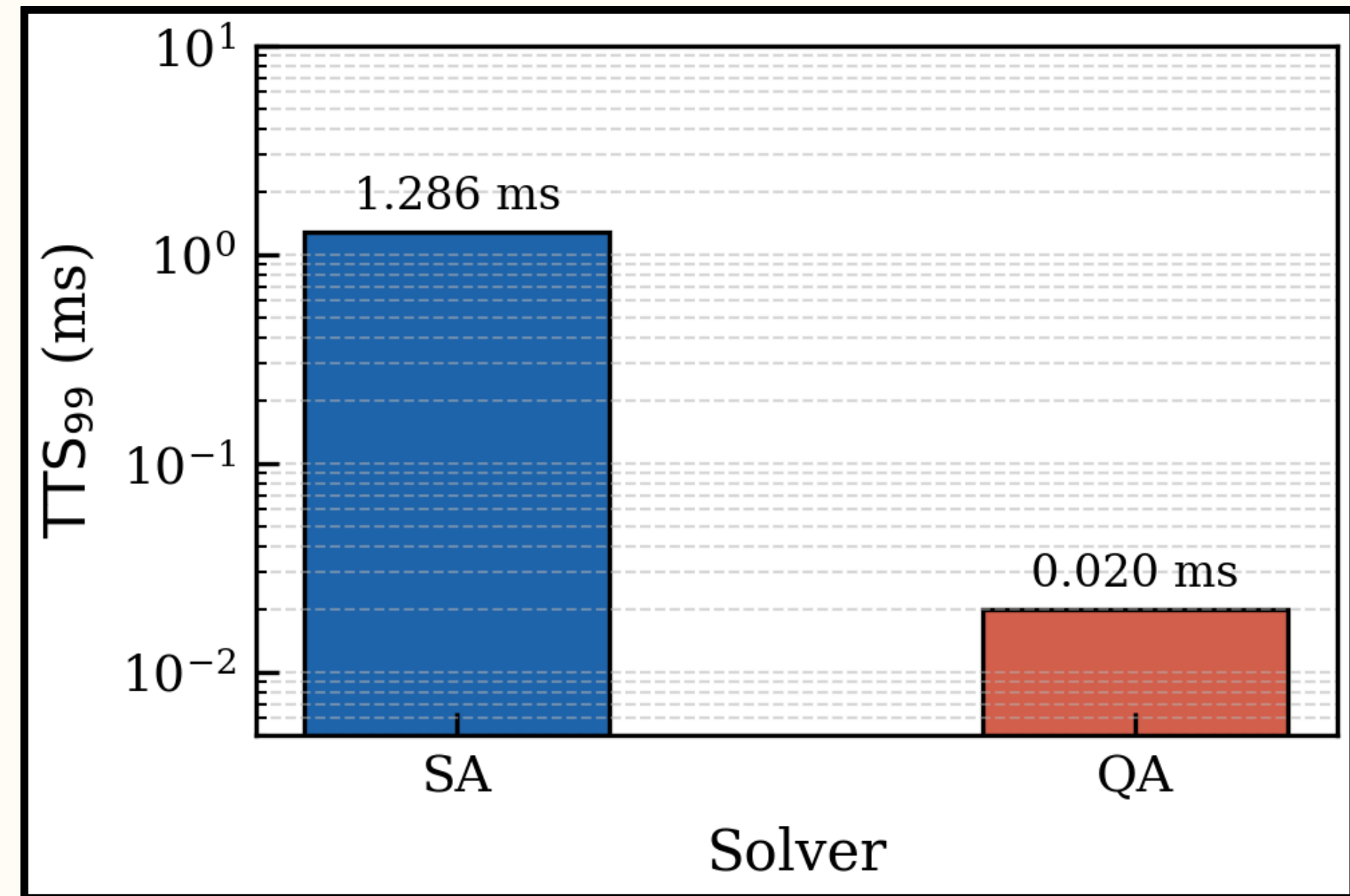


Figure 10: TTS₉₉ per QUBO solve for SA (measured) and QA (projected D-Wave, $\tau = 20 \mu\text{s}$) — log scale

Source: Authors

Our Present Work: Second Paper Drafted



Source: Authors

- **The Problem**
 - Which AP-MP node pairs should a device-free localization system activate under separate AP and MP hardware budgets?
- **Our Contribution**
 - A QUBO formulation that jointly rewards link importance and penalizes inter-link redundancy
 - A formal monotonicity proof guaranteeing that binary search over α finds the optimal feasible deployment in exactly 7 iterations for $\varepsilon = 0.01$
 - QA achieves 2.66 m mean error with a projected 129× per-solve speedup over SA
- **Why It Advances the Field**
 - First penalty-free budget enforcement mechanism for two simultaneous hardware constraints in DFL
 - Formulation generalizes to any link-based DFL system
- **Target Venues**
 - **Primary:** IEEE INFOCOM | **Fallback:** IEEE LCN

Conclusion

- **We formulated AP-MP node placement in DFL as a QUBO that jointly rewards link importance and penalizes inter-link redundancy**
- **A formal monotonicity proof guarantees that binary search over α finds the optimal feasible deployment $\rightarrow$ no penalty coefficient needed**
- **Over 100 runs: QA achieves lower mean localization error than SA with a $129\times$ per-solve speedup**
- **The formulation generalizes to any link-based DFL system that produces RSS fingerprints $\rightarrow$ not specific to RTI**

Future Work

- **Validate on physical D-Wave hardware**
 - All QA speedup numbers in this work are projected from D-Wave specifications
 - Running the binary search on real hardware would confirm the $129\times$ and $5,839\times$ claims under actual noise and readout overhead
- **Node-level QUBO representation**
 - The current formulation uses directed AP-MP pairs as decision variables, giving a matrix of size $P = N(N-1) = 90$ for $N = 10$
 - Reformulating over node-level binary variables would reduce the matrix to size N , making the approach scale to much larger candidate pools without hitting D-Wave embedding limits
- **Online and adaptive deployments**
 - The current system selects a deployment once offline before localization begins
 - Extending to online settings where nodes can be reconfigured in response to observed localization error

AI Use Disclosure

The following AI tools were used in the preparation of this thesis and presentation in accordance with AUC academic integrity policy.

Writing Assistance

- **Claude (Anthropic)** was used to improve the clarity, structure, and flow of written sections in the research paper, including the abstract, introduction, related work, and conclusion
- All technical content, results, claims, and mathematical proofs are entirely the work of the authors

Code and Experiments

- **Claude Code** was used to assist in generating experimental code and running simulations.

Presentation

- **Claude (Anthropic)** was used to help structure slide content and refine bullet point phrasing

Literature and References

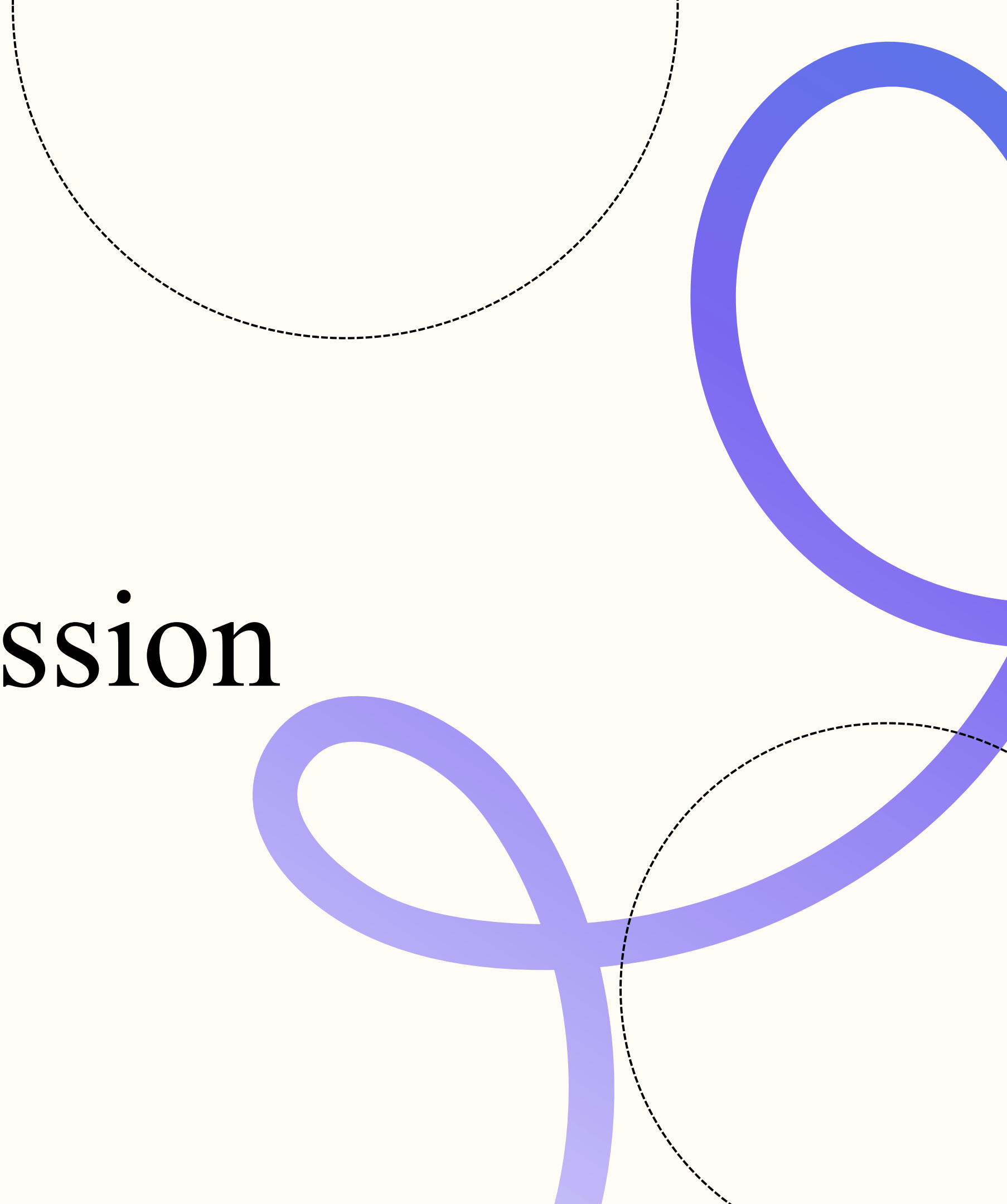
- **Perplexity** was used to help find, summarize, and generate references
- All 28 citations were verified manually by the authors

Figures' Sources

- [F1] H. Huang, H. Zhao, X. Li, S. Ding, L. Zhao, and Z. Li, "An Accurate and Efficient Device-Free Localization Approach Based on Sparse Coding in Subspace," *IEEE Access*, vol. 6, pp. 61782–61799, 2018. DOI: 10.1109/ACCESS.2018.2876034.
- [F2 and F5] Google. (2026). Gemini [AI image generator]. Diagrams generated May 2026.
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- [F3] Interactive HTML animation generated using Claude Code (Anthropic), May 2026.
- [F4 - F5 and F7 - F10] Authors.

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Thank You!

Questions & Discussion

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